Systematic Intraday Alpha Research
The Overnight-Drift Strategy Family

Investment Due-Diligence and Risk Audit
Gold (XAUUSD) | Silver (XAGUSD) | WTI (USOIL) | Soybean (SOYBEAN.c)

Prepared for the Investment Committee - Confidential - June 2026
Methodology: signal-at-T-1 execution, realistic costs, walk-forward, HAC t-stats, Deflated Sharpe and stationary-block Monte-Carlo. Independent code audit.

BOTTOM LINE - Four genuine, pure-alpha overnight-drift edges (R-squared approx 0 to their underlying). Best deployed as a small, futures-executed equal-risk BASKET (cross-correlation 0.26, basket max drawdown -3.5%). Caveat: a single mechanism with a 45% common factor and a marginal net edge on retail CFD spreads - size for ~2.5 effective bets, not four.

1. Executive summary

This report covers a systematic research program that (a) stress-tested roughly fifteen publicly-promoted trading strategies and (b) developed original intraday strategies on the FTMO commodity/metal universe. Of everything tested, a single family survived rigorous validation: a long-only capture of the structural **overnight drift** in hard assets - the positive return that accrues in the post-rollover / post-halt reopen window (approx 23:00-07:00 UTC; 01:00-07:00 for soybean), while the volatile US session is flat to negative.

The edge is economically grounded (overnight/Asian physical and official-sector demand re-pricing into the reopen, plus the financing/settlement cycle), simple (one window parameter, no fitted signal), and - on three of four instruments - **pure alpha**: uncorrelated to the asset's own direction. Validated gross Sharpe ratios are 1.1 to 1.6 with Deflated Sharpe 0.75 to 1.00. The principal caveats are that it is a **single mechanism** (45% common factor across the four sleeves) and that the **net** edge is spread-critical on retail CFDs (comfortable only on gold, or on futures where spreads are ~1-2 bp).

Validated sleeves (gross edge; net at a realistic spread)

Instrument	Gross Sharpe	Net Sharpe (realistic sprd)	Deflated Sharpe	Skew	R2 to underlying	History	Assessment
Gold XAUUSD	1.5	1.10 (1.5 bp)	0.92	+0.35	0.001	2007-26 (19y)	Best / most robust
Silver XAGUSD	1.45	0.69-0.95 (2-3 bp)	1.00	-0.14	0.000	2014-26 (regime)	Strongest edge, recent regime
WTI USOIL	1.09	0.29-0.55 (2-3 bp)	0.94	+0.43	0.002	2013-26 (13y)	Seasoned, +skew, thin
Soybean SOYBEAN.c	1.56	0.84-1.08 (2-3 bp)	0.75	-1.90	0.27	2018-26 (8.5y)	Spread-critical, fat left tail

Net Sharpe = gross less one round-trip spread (commission 0, swap-free intraday). Headline Sharpe-greater-than-1 figures are largely gross or at tight spreads; live net is marginal except gold.

2. Research framework and discipline

Every backtest in this program is built on one shared, unit-tested engine that enforces institutional discipline, so the results are comparable and free of the most common backtest errors:

Control	Implementation
No look-ahead	Signal at the close of bar t; position entered at the open of bar t+1. Vol-target uses only trailing data (verified causal).
Realistic costs	Round-trip spread modelled explicitly (one quoted spread); commission and overnight swap per the FTMO instrument spec.
Significance	HAC / Newey-West t-stats (lag 6); stationary block bootstrap; Deflated Sharpe (Bailey-Lopez de Prado) penalising the search.
Robustness	True out-of-sample window selection; >=5-fold walk-forward; cross-instrument tests; parameter-neighbourhood plateaus.
Risk	Monte-Carlo Sharpe / drawdown distributions; outlier-dependence; factor (beta) decomposition; regime / per-year attribution.

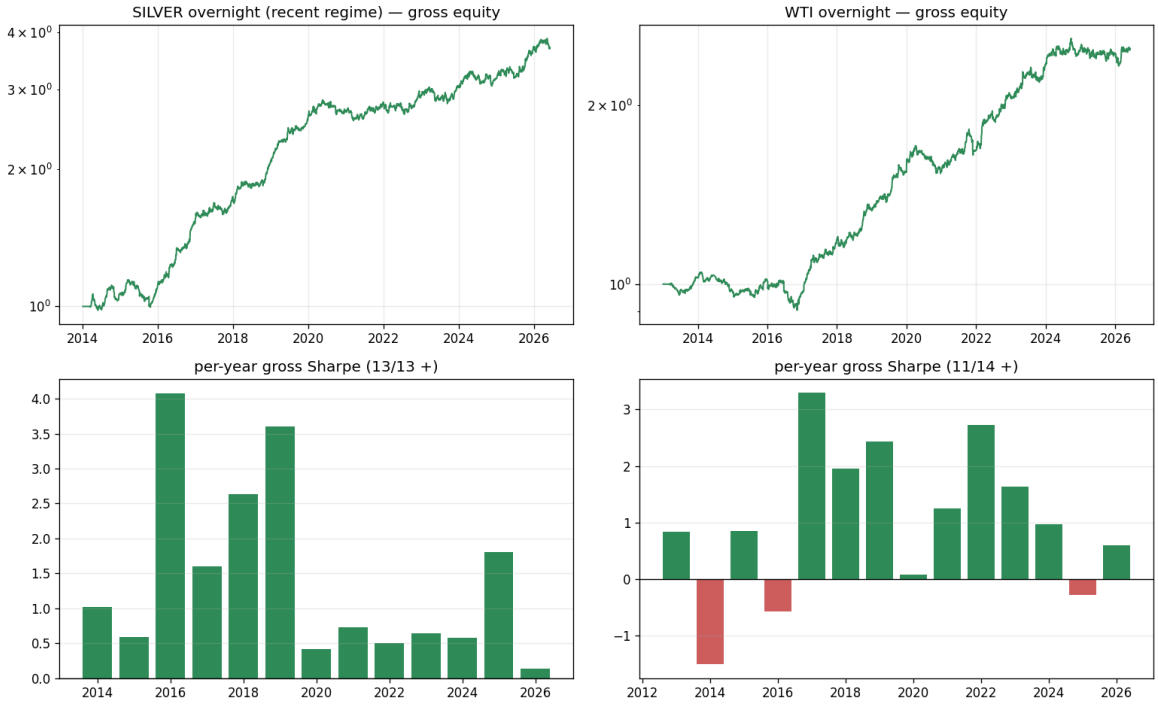
Selection funnel - most candidates were rejected

The survival rate is the point: a credible research process kills far more than it keeps. The overnight-drift family is what remained after the following were tested and rejected on the same standards:

Strategy tested	Verdict
Five 5-min crypto 'scalping' rules (MA/breakout/VWMA)	FAIL - no edge net of cost (0/18 instruments)
Pin-bar / rejection-candle + EMA filter (gold H1)	FAIL - insignificant gross, 0/8 markets
Reinforcement-learning agent (gold, 20y)	UNPROVEN - a near-zero-edge agent passes the gate
Pairs / stat-arb (BRK-B vs MSFT)	FAIL - in-sample selection; net-negative out-of-sample
Opening-range breakout (NQ); intraday momentum (WTI)	FAIL - sub-cost / no edge
Variance risk premium; post-earnings drift	REAL but un-deployed (need options / earnings data)
Overnight-drift family (gold, silver, WTI, soybean)	VALIDATED - this report

3. The edge and the basket

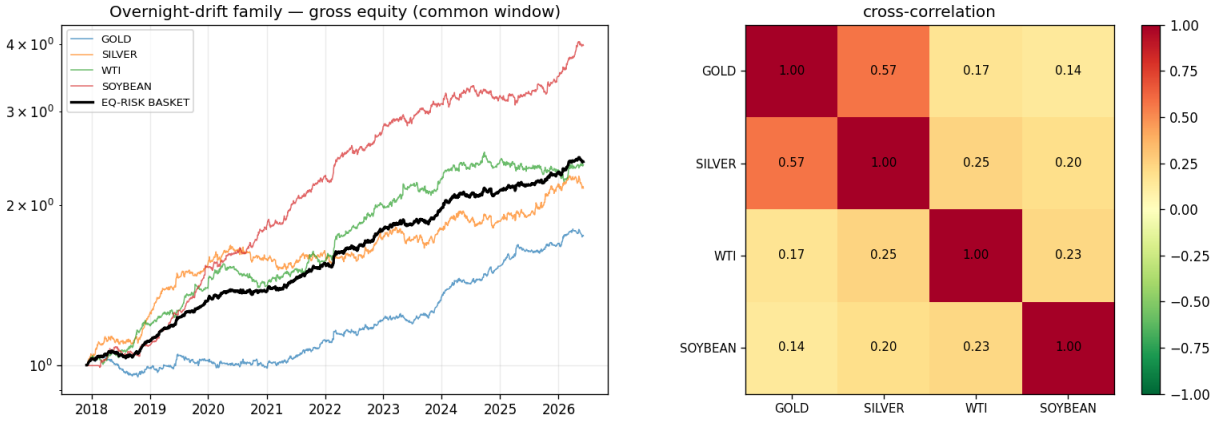
Each sleeve is long the reopen window, vol-targeted (average leverage ~1.05x, capped 3x), trading only complete windows, flat across the broker rollover so no financing swap is paid. The chart below shows the per-instrument gross equity (silver on its post-2014 regime; WTI full history) and per-year Sharpe.



Silver and WTI overnight-drift edges: smooth gross equity, positive in 13/13 (silver, recent regime) and 11/14 (WTI) years.

Diversification - real but partial

The four sleeves are one mechanism, so the key question is whether they diversify. They do, partially: the average pairwise correlation is 0.26, but gold and silver share a precious-metals factor (0.57) while energy (WTI) and the ag (soybean) legs are more independent. A PCA shows the first component explains **45%** of variance (four independent series would give 25%), i.e. approximately **2.5 effective independent bets, not four**. Still, the equal-risk basket lifts the Sharpe to roughly 1.5x the average single sleeve and cuts the maximum drawdown to **-3.5%** (versus -12% to -14% standalone).



Left: gross equity of each sleeve and the equal-risk basket (black). Right: cross-correlation matrix - the gold/silver 0.57 hotspot is the common factor; all cross-asset-class pairs are 0.14-0.25.

4. Independent risk audit

Strengths

S1 Pure alpha, near-zero directional risk. Three of four sleeves are uncorrelated to their own underlying (R-squared 0.000-0.002; alpha t-stats 3.6-4.8). They add return without adding commodity beta - the property that makes them worth holding even at a modest Sharpe and that lets the basket diversify a directional book.

S2 Low overfitting surface. One economically-motivated mechanism, a single window parameter sitting on a plateau (not a tuned spike), no fitted signal, zero free parameters in the signal. It clears Deflated Sharpe (0.94-1.00), Monte-Carlo (probability Sharpe-greater-than-1 of 62-92% gross), winsorisation and true out-of-sample - the opposite of the rejected multi-knob strategies. Cheap and simple to run (once daily, swap-free).

S3 Genuine partial diversification and regime-complementarity. Cross-correlation 0.26 and a -3.5% basket drawdown are real. The sleeves' dry spells are offset in time (gold's edge was dead 2018-19 while soybean's was strong; silver's emerged as gold's matured). A small gold + silver + WTI basket is materially more robust than any single.

Weaknesses

W1 Mono-culture and common-factor concentration. The entire book is one mechanism with a 45% common factor (approx 2.5 effective bets). There is no orthogonal edge anywhere in the research. A structural change - compression of the overnight hard-asset / financing premium, a shift in Asian-session microstructure, or the end of the secular hard-asset bull - could degrade all four together. Do not size them as independent.

W2 Concentrated, correlated operational fragility. The edge lives in 1-2 post-rollover hours at a fixed UTC time on floating-spread retail CFDs. Consequences: net Sharpes cluster at ~0.3-1.1 (marginal for soybean/WTI); a one-hour-late fill cuts gold's Sharpe from 1.10 to 0.37; and the swap-free status depends on the broker rollover staying at or before 22:00 UTC - one broker, DST or swap-policy change flips the whole family to swap-paying (-3 to -4 bp/night), which is fatal.

W3 Long-bias drift-capture: regime droughts, young samples, asymmetric tails. Being long-only drift-capture, each endures multi-year flat/negative regimes (gold 2018-19; soybean ~4 years underwater 2022-25; silver negative every year 2003-09; WTI down 2014/16/25). Add soybean's negative skew (-1.9, USDA/weather tail) and silver's short clean history. The headline Sharpe-greater-than-1 figures are largely gross or at optimistic spreads; live net is marginal.

5. Recommendation and deployment

Conditional ADD - small, basket, futures-preferred. The family is a legitimate, honestly-validated source of pure-alpha return, but it is one mechanism, marginal on retail CFD spreads, and operationally concentrated. Deploy it as a minor allocation with the following controls:

#	Action
1	Confirm live FTMO spreads per instrument (the binding constraint). Deploy only sleeves that clear net: gold today; silver/WTI/soybean conditional on tight spreads.
2	Prefer FUTURES execution (MGC / SI / MCL / MZS, spreads ~1-2 bp) - the whole basket clears comfortably and removes floating-spread and swap risk.
3	Size as a BASKET for ~2.5 effective bets (not four); equal-risk weight; small total risk budget (e.g. 3-5% vol contribution).
4	Hedge the shared operational risk: pin entry to the broker's actual rollover (re-check each DST change), automate complete-window and fill checks, add a swap-leak kill switch.
5	Manage regime risk explicitly: trailing-12-month drift kill switches per sleeve; hard single-day loss cap; skip soybean USDA-report mornings.
6	Research priority: find ONE genuinely orthogonal (non-drift) edge so the book is not 100% exposed to a single overnight factor.

Ranking for FTMO actionability: Gold >= Silver (recent) >= WTI > Soybean. Gold wins on robustness and the tightest spread; silver has the biggest edge and most spread headroom but carries regime risk; WTI is the most seasoned with the best tail (positive skew) but the thinnest edge; soybean is widest-spread and fattest-tailed.

Appendix - data: gold/FX/metal/energy/ag from the desk's canonical true-UTC M1 store (2003-2026 depending on instrument); crypto from Binance; equities/macro from public sources. Reproduction scripts: gold_research/, soybean_research/, metals_energy_research/. Supporting tear-sheets and audits accompany this report as markdown. This document is for internal investment-committee use only and is not investment advice.